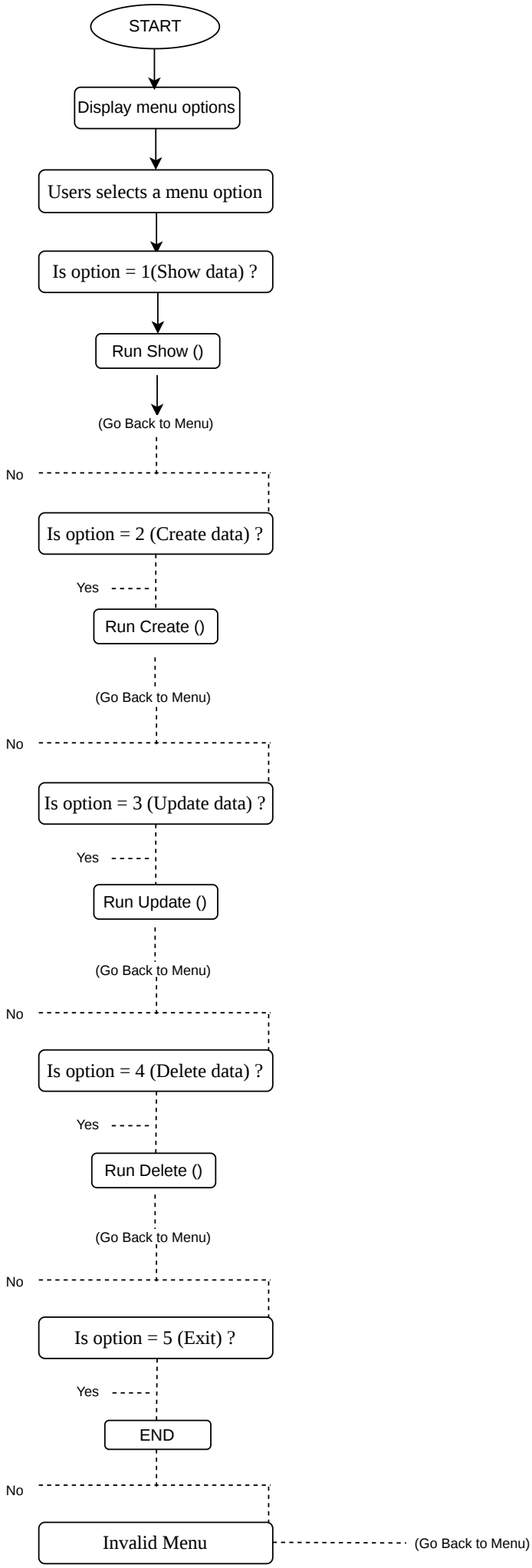
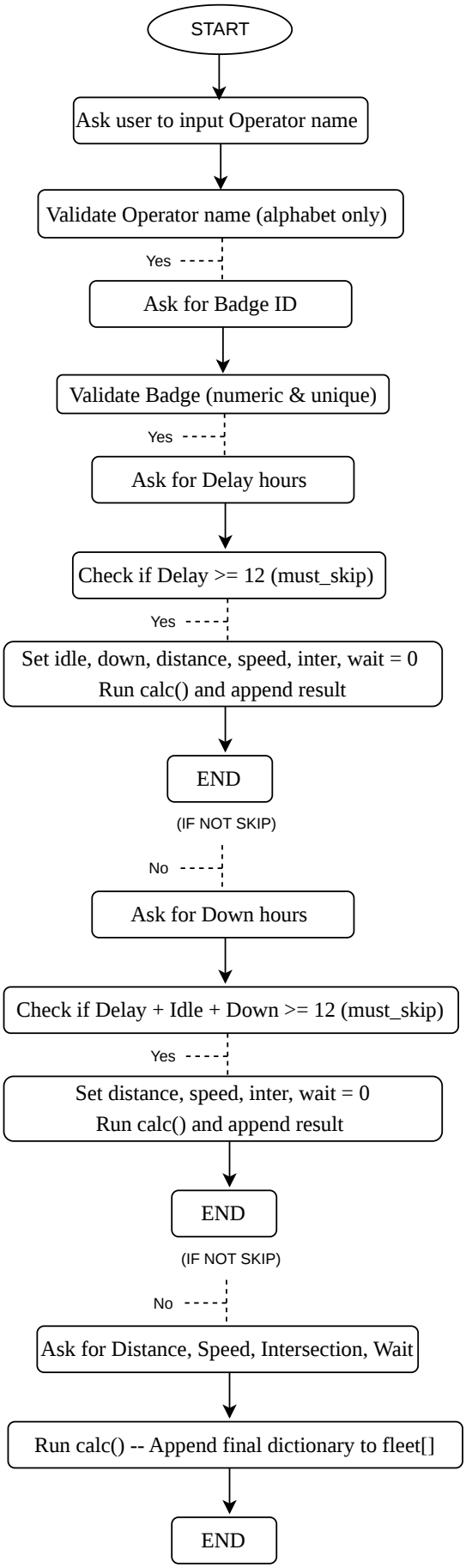
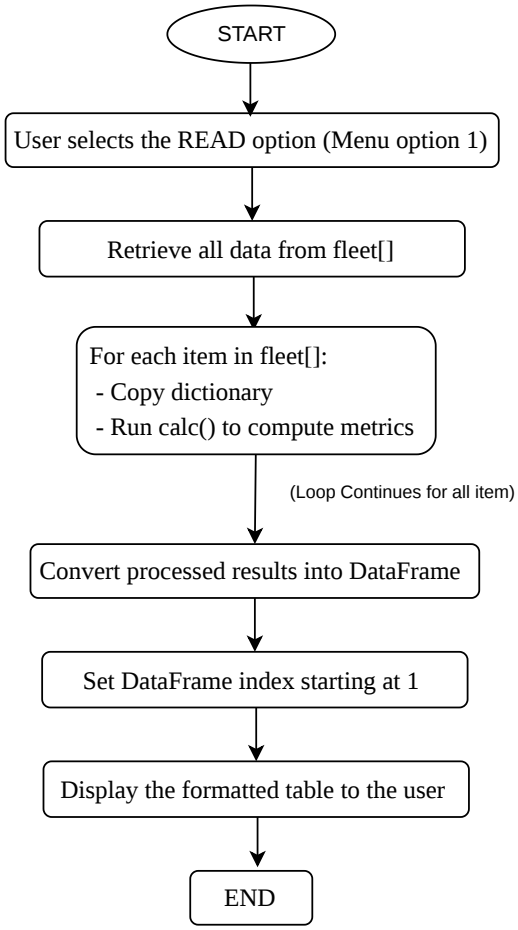
MAIN MENU


START
Display menu options
Users selects a menu option
Is option = 1(Show data) ?
Run Show ()
(Go Back to Menu)
No
Is option = 2 (Create data) ?
Yes
Run Create ()
(Go Back to Menu)
No
Is option = 3 (Update data) ?
Yes
Run Update ()
(Go Back to Menu)
No
Is option = 4 (Delete data) ?
Yes
Run Delete ()
(Go Back to Menu)
No
Is option = 5 (Exit) ?
Yes
END
No
Invalid Menu
(Go Back to Menu)

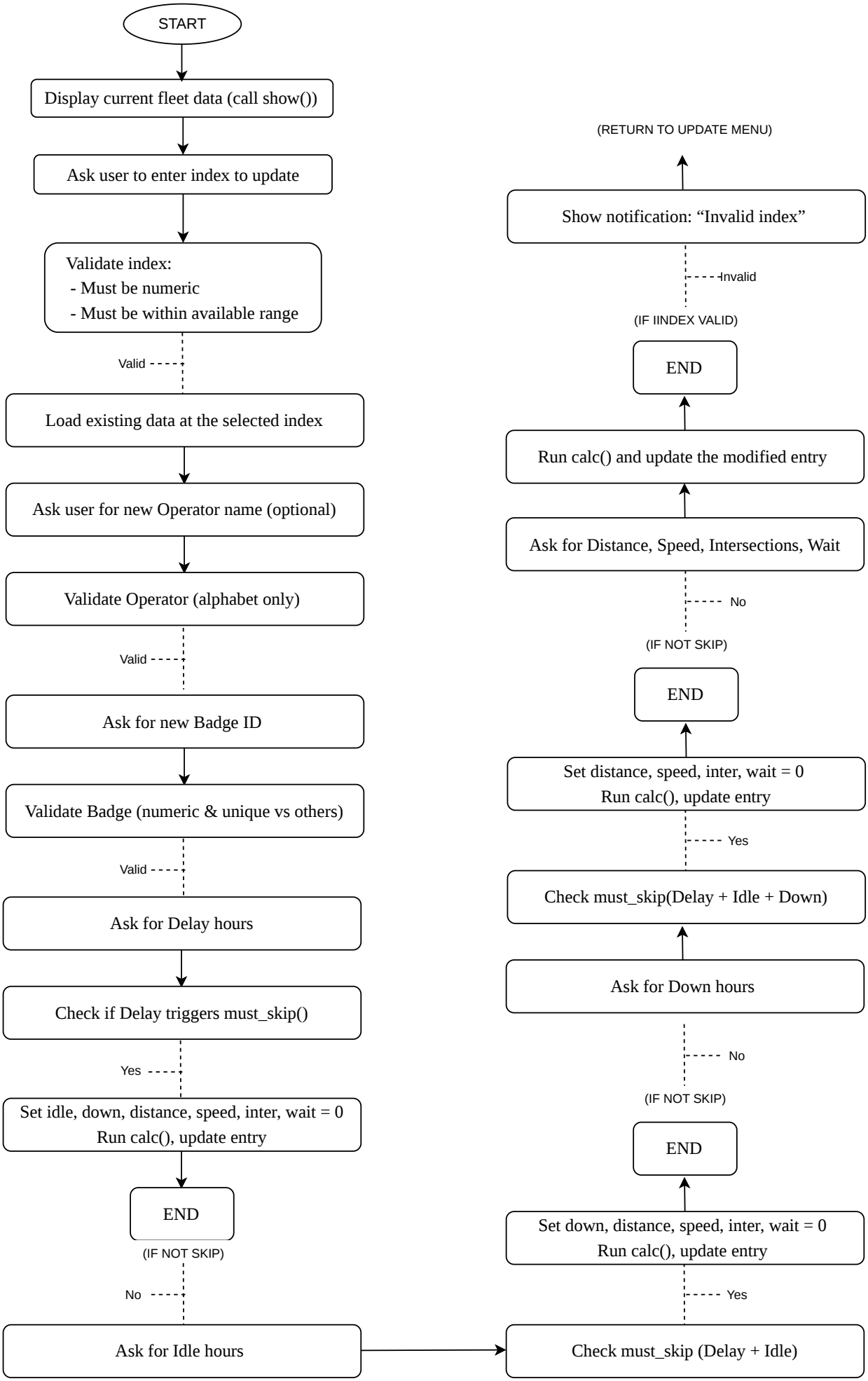
CREATE MENU


START
Ask user to input Operator name
Validate Operator name (alphabet only)
Yes
Ask for Badge ID
Validate Badge (numeric & unique)
Yes
Ask for Delay hours
Check if Delay >= 12 (must_skip)
Yes
Set idle, down, distance, speed, inter, wait = 0
Run calc() and append result
END
(IF NOT SKIP)
No
Ask for Down hours
Check if Delay + Idle + Down >= 12 (must_skip)
Yes
Set distance, speed, inter, wait = 0
Run calc() and append result
END
(IF NOT SKIP)
No
Ask for Distance, Speed, Intersection, Wait
Run calc() -- Append final dictionary to fleet[]
END

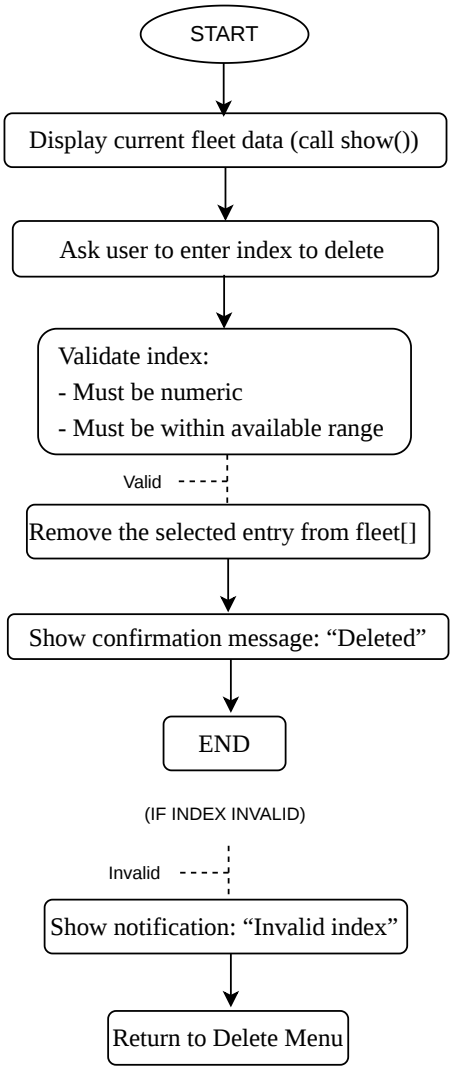
READ MENU


START
User selects the READ option (Menu option 1)
Retrieve all data from fleet[]
For each item in fleet[]:
- Copy dictionary
- Run calc() to compute metrics
(Loop Continues for all item)
Convert processed results into DataFrame
Set DataFrame index starting at 1
Display the formatted table to the user
END

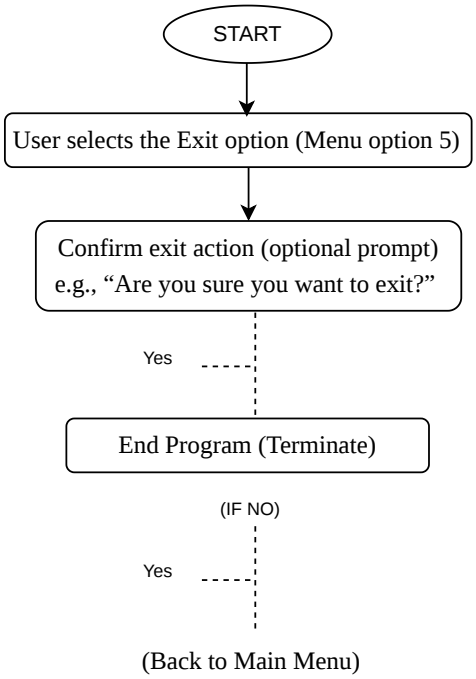
UPDATE MENU


START
Display current fleet data (call show())
Ask user to enter index to update
Validate index:
- Must be numeric
- Must be within available range
Valid
Load existing data at the selected index
Ask user for new Operator name (optional)
Validate Operator (alphabet only)
Valid
Ask for new Badge ID
Validate Badge (numeric & unique vs others)
Valid
Ask for Delay hours
Check if Delay triggers must_skip()
Yes
Set idle, down, distance, speed, inter, wait = 0
Run calc(), update entry
END
(IF NOT SKIP)
No
Ask for Idle hours
Check must_skip (Delay + Idle)
Yes
Set down, distance, speed, inter, wait = 0
Run calc(), update entry
END
(IF NOT SKIP)
No
Ask for Down hours
Check must_skip(Delay + Idle + Down)
Yes
Set distance, speed, inter, wait = 0
Run calc(), update entry
END
(IF NOT SKIP)
No
Ask for Distance, Speed, Intersections, Wait
Run calc() and update the modified entry
END
(IF IINDEX VALID)
Invalid
Show notification: “Invalid index”
(RETURN TO UPDATE MENU)

DELETE MENU


START
Display current fleet data (call show())
Ask user to enter index to delete
Validate index:
- Must be numeric
- Must be within available range
Valid
Remove the selected entry from fleet[]
Show confirmation message: “Deleted”
END
(IF INDEX INVALID)
Invalid
Show notification: “Invalid index”
Return to Delete Menu

EXIT MENU


START
User selects the Exit option (Menu option 5)
Confirm exit action (optional prompt)
e.g., “Are you sure you want to exit?”
Yes
End Program (Terminate)
(IF NO)
Yes
(Back to Main Menu)